

Definition (Diagramatic invariant). $I : \text{knot} \rightarrow S$ with S set such that $I(K_1) = I(K_2)$ if K_1 and K_2 are isotopic.

Example. The crossing at a cross $\varepsilon(c) \in \{+1, -1\}$. The linking number of a link $L = K_1 \cup K_2$ is an invariant defined as

$$lk(K_1, K_2) = \frac{1}{2} \sum_{c \in K_1 \cap K_2} \varepsilon(c).$$

Definition (Fox p -coloring). Let D a knot diagram with arcs $x_1, \dots, x_m$. A Fox p -coloring of D is an assignment of colors

$$x_i \mapsto a_i \in \mathbb{Z}_p$$

such that at each crossing, we have a, c underarcs and b overarc, then $2b \equiv a + c \pmod{p}$.

We have a homogeneous system of equations $C_D x = 0$

Proposition. The dimension of

$$Col_p(D) = \ker(C_D)$$

(hence, the number of Fox p -colorings) is a knot invariant.

Definition (Knot group). Let K be a knot. The knot group of K is $\pi_1(S^3 \setminus K)$.

Definition. $N(K)$ tubular nbh of K .

Definition (Meridian and longitude). They generate $\pi_1(\partial N(K)) \cong \mathbb{Z}^2$.

Theorem (Wirtinger presentation). g_i generator for each arc x_i . Then the knot group has presentation

$$G_K = \langle g_1, \dots, g_m \mid r_1, \dots, r_n \rangle$$

where $r_j : g_k = g_i g_j g_i^{-1}$ are wirtinger relations at each crossing.

Definition. $C_D^{i,j}$ deleting the i -th row and j -th column of C_D . Then

$$\det(K) = \left| \det(C_D^{i,j}) \right| = |\Delta_K(-1)| = |H_1(\Sigma_2(K))|$$

Example. *Trifoil*

$$C_D = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Theorem. K admits a non-trivial Fox p -coloring if and only if $p \mid \det(K)$.

Definition (Writhe). Not a knot invariant (fails Reidemeister I)

$$w(D) = \sum_c \varepsilon(c)$$

Definition (Framed knot). A knot K equipped with $v(x)$ a normal vector field. Self-linking number

$$sl(K) = lk(K, K_v = \{x + \varepsilon v(x) : x \in K\})$$

1 Seifert and Alexander

Definition (Seifert surface). For a knot K , a Seifert surface is a compact connected orientable surface F such that $\partial F = K$.

Proposition. Seifert's algorithm: *TODO*

Definition (Euler characteristic).

$$\chi(F) = s(D) - c(D) = 1 - 2g(F)$$

where $s(D)$ is the number of Seifert circles and $c(D)$ is the number of crossings.

Proposition. $K_1 \# K_2$ induces $F_1 \natural F_2$ and

$$g(F_1 \natural F_2) = g(F_1) + g(F_2), \quad \chi(F_1 \natural F_2) = \chi(F_1) + \chi(F_2) - 1, \quad g(F_1 \natural F_2) = g(F_1) + g(F_2)$$